

# Species-Aware Module Review: EF-P

## Translation Stall Rescue in *Pseudomonas putida* KT2440 (UniPathway UPA00345)

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**Target:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556)

**Bucket:** unipathway:UPA00345 — EF-P-dependent stimulation of peptide-bond formation / rescue of stalled cytosolic ribosomes **Anchor gene:** efp = PP\_1858 = UniProtKB **Q88LS0** (EFP\_PSEPK) **Review completed:** Iteration 3 of 3 (final)

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### 1. Executive summary

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The single-gene UPA00345 module for *P. putida* KT2440 is **SATISFIED and COVERED** by the sole candidate gene **efp (PP\_1858, Q88LS0)**. This is a canonical, universally conserved translation elongation factor P that rescues ribosomes stalled at polyproline (and related) motifs by stabilizing P-site peptidyl-tRNA and promoting peptide-bond formation. Evidence is **direct for the target strain** at the sequence/annotation level (reviewed SwissProt entry, HAMAP MF\_00141, Arg32 rhamnosylation feature) and mechanistically strong from genus-level and structural studies.

KT2440 EF-P uses the **pseudomonad activation strategy: Arg32 rhamnosylation by EarP** (PP\_1857, immediately adjacent), using **dTDP-L-rhamnose** supplied by the complete *rml/rfb* nucleotide-sugar cluster (PP\_1782–1785; RmlA–D). These activation genes are **real and present** but are correctly **outside the UPA00345 bucket**; they belong to separate glycosyltransferase / nucleotide-sugar modules. The *E. coli*-style EpmA/EpmB/EpmC  $\beta$ -lysylation route is **not expected** in this organism (EF-P carries Arg, not Lys, at position 32). KT2440 has **no EfpL/YeiP paralog**, so there is no paralog ambiguity and the single-gene module scope is correct.

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## 2. Target-organism pathway definition

- **Process included:** EF-P-dependent stimulation of peptidyl transferase activity to rescue elongating 70S ribosomes stalled at consecutive prolines (PPP/PPG and related XPP/PPX motifs). EF-P binds between the P- and E-sites, contacts the CCA end of P-site tRNA, and enforces a nascent-chain geometry compatible with peptide-bond formation

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- **Keep separate (neighboring processes, NOT this bucket):**
- **EarP-catalyzed EF-P Arg-rhamnosylation** (post-translational activation; a glycosyltransferase reaction) — separate module/GO.
- **dTDP-L-rhamnose biosynthesis** (*rmlABCD* / *rfbABCD*; nucleotide-sugar metabolism, also feeds LPS O-antigen) — separate module.
- **General polypeptide elongation** (UniProt PATHWAY "Protein biosynthesis; polypeptide chain elongation") — the broad overview map; UPA00345 is the specialized stall-rescue sub-process, not general elongation.
- **Ribosome rescue by tmRNA/ArfA/ArfB** (rescue of *truncated-mRNA*-stalled ribosomes) — a mechanistically distinct rescue system; do not merge.
- **Alternate names / database definitions:** EF-P (efp); eukaryotic/archaeal functional analog eIF5A/aIF5A ("EF5"); UniProt keyword "Elongation factor"; HAMAP rule MF\_00141; the modified-arginine subfamily is the "PGKGP/Arg32" type of EF-P.

## 3. Expected step model

UPA00345 is a **single-step** bucket in the metadata, and that matches the biology for KT2440:

| Step | Description                                                                                    | Expected gene | Status in KT2440                  |
|------|------------------------------------------------------------------------------------------------|---------------|-----------------------------------|
| S1   | EF-P-dependent rescue of polyproline-stalled ribosomes / stimulation of peptide-bond formation | efp           | <b>Covered</b> (PP_1858 / Q88LS0) |

Activation context (adjacent modules, listed for curator awareness, **not** UPA00345 steps):

| Context step              | Gene(s) in KT2440                                                                                                     | Note                                                          |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| EF-P Arg32 rhamnosylation | earP / PP_1857 (Q88LS1)                                                                                               | Adjacent to efp; experimental EC 2.4.1.- rhamnosyltransferase |
| dTDP-L-rhamnose supply    | Complete Rml cluster PP_1782–1785: rmlA/rfbA (PP_1783), rmlB/rffG (PP_1785), rmlC/rfbC (PP_1782), rmlD/rfbD (PP_1784) | Nucleotide-sugar / O-antigen module; supplies EarP substrate  |

## 4. Candidate genes and evidence

### efp — PP\_1858 — Q88LS0 (EFP\_PSEPK) — high confidence, COVERED

- **Role:** Elongation factor P; rescues polyproline-stalled ribosomes and stimulates peptide-bond synthesis on 70S ribosomes (UniProt FUNCTION; HAMAP MF\_00141).
- **Evidence type:** Direct SwissProt annotation for strain KT2440; 189 aa; "Belongs to the elongation factor P family." Sequence position 32 = **Arg**, carrying UniProt feature "**Glycosylation 32..32 N-alpha-linked (Rha) arginine**" and PTM note "arginine rhamnosylation is required for EF-P function and rescue of polyproline stalled ribosomes." Function is HAMAP-rule/homology inferred; the PTM/site is supported by genus-level experiments

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- **Curation caveats:** None that undermine the assignment. It is the **only** EF-P-family protein in the KT2440 proteome (no over-propagation risk, no paralog to disambiguate). GO/pathway mappings are specific and appropriate.

### earP — PP\_1857 — Q88LS1 (EARP\_PSEPK) — activation context (out of bucket)

- **Role:** Protein-arginine rhamnosyltransferase; transfers rhamnose onto EF-P Arg32 (required for EF-P activity). Experimental support ECO:0000269

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- **Curation caveat:** UniProt FUNCTION text says the target is "'Lys-32'" — this is a **residue mislabel**; the modified residue is **Arg32** (efp position 32 = R; feature = arginine glycosylation). Flag for UniProt correction. Do **not** place earP inside UPA00345.

## 5. Gaps, ambiguities, and likely over-annotations

- **No gap** in the UPA00345 core: the single expected step is present with a specific, reviewed gene.
- **Not expected in target taxon:** the *E. coli* EF-P  $\beta$ -lysylation activation genes **epmA (poxA/genX)**, **epmB (yjeK)**, **epmC**. A UniProt proteome search of taxon 160488 returned **zero** hits for any of these (genomic absence, not just inference). KT2440 EF-P has **Arg** (not Lys) at the modification site (loop G31-**R32**-N33) and uses the EarP route instead. Any epmA/B/C annotation transferred to KT2440 should be treated as **not\_expected / mis-transfer**.
- **Activation substrate fully supplied:** the complete dTDP-L-rhamnose cluster **PP\_1782–1785** (RmlA/RmlB/RmlC/RmlD) is present, so EarP's sugar-nucleotide donor is genomically guaranteed.
- **Paralog ambiguity:** none. Unlike *E. coli* (which has EfpL/YeiP,

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KT2440 encodes a single EF-P. No risk of splitting the assignment between paralogs.

- **Over-annotation to watch:** broad "translation elongation factor activity" / GO:0003746 could be applied loosely; for UPA00345 the specific, evidenced function (polyproline stall rescue) is what matters and is well supported here.
- **Boundary correctness:** the generic module boundary (single efp gene, with EarP/ rhamnose flagged as activation context, not members) is **correct** for this organism; no revision needed.

## 6. Module and GO-curation recommendations

- **S1 (EF-P stall rescue):** mark **COVERED** — efp / PP\_1858 / Q88LS0.

- **Module scope: keep single-gene;** do **not** fold earP or the dTDP-L-rhamnose (*rml/rfb*) genes into UPA00345. Represent EarP activation and rhamnose supply as **separate linked modules** if cross-references are desired.
- **not\_expected\_in\_target\_taxon:** EpmA/EpmB/EpmC ( $\beta$ -lysine) activation route.
- **module\_needs\_revision:** No.
- **New GO term / document requests:** None required for UPA00345 itself. Optionally request/verify a curated cross-reference linking EF-P (GO:0043022/GO:0003746, ribosome-associated stall rescue) to the EarP rhamnosylation activation module, and file a **UniProt correction** for the Q88LS1 "Lys-32"  $\rightarrow$  "Arg-32" residue text.

## 7. Genes to promote to full review

1. **efp (PP\_1858, Q88LS0)** — promote to full `fetch-gene` review as the module anchor (highest priority; the covered step).
2. **earP (PP\_1857, Q88LS1)** — optional secondary promotion as the obligate activation partner (adjacent gene, experimental evidence, contains the annotation typo to fix). Not a UPA00345 member but essential to interpret EF-P functionality in this taxon.

## 8. Key references

- Lassak J. *et al.* (2015) *Arginine-rhamnosylation as new strategy to activate translation elongation factor P*. **Nat Chem Biol** — PMID 25686373 (corrigendum [P 25785429](#)).
- Krafczyk R. *et al.* (2017) *Structural Basis for EarP-Mediated Arginine Glycosylation of EF-P*. — PMID 28951478. Rhamnosylation on Arg32; cited by KT2440 UniProt entries.
- Huter P. *et al.* (2017) *Structural Basis for Polyproline-Mediated Ribosome Stalling and Rescue by EF-P*. — PMID 29100052. Mechanism of stall rescue.
- Yanagisawa T. *et al.* (2016) *N. meningitidis EF-P and its active-site Arg are essential...* — PMID 26840407. EF-P/Arg-modification dispensable in *P. aeruginosa*/*S. oneidensis*.

- Lassak J., Wilson D.N., Jung K. (2016) *Stall no more at polyproline stretches...* (review) — PMID **26416626**.
  - Sieber A. *et al.* (2024) *EF-P and its paralog EfpL (YeiP)...* — PMID **39622818**. EfpL/YeiP paralog (absent in KT2440).
  - UniProtKB: **Q88LS0** (EFP\_PSEPK), **Q88LS1** (EARP\_PSEPK); HAMAP rule MF\_00141.
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## Evidence provenance summary

- **Direct for KT2440:** UniProt reviewed annotations (efp Q88LS0, earP Q88LS1), efp Arg32 sequence (loop G31-R32-N33) + rhamnosylation feature, gene adjacency PP\_1857/PP\_1858, complete *rml* cluster PP\_1782–1785, and genomic **absence** of epmA/B/C.
  - **Genus/strong transfer (*P. aeruginosa*, *Pseudomonas*):** EarP rhamnosylation chemistry, requirement for EF-P function, non-essentiality.
  - **Homology/rule-based:** core EF-P peptide-bond-formation function (HAMAP MF\_00141).
  - **Open questions:** direct experimental demonstration of polyproline-stall rescue specifically in KT2440 (vs. *P. aeruginosa*) is lacking; whether EF-P/EarP loss produces a measurable fitness/proteome phenotype in KT2440's saprophytic lifestyle is untested; correct the UniProt Q88LS1 "Lys-32" text to "Arg-32". (The *rmlB* locus is now confirmed as rffG/PP\_1785.)
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